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OMEGA SERIES 4.3

TSMVEFT: A Two-Sector Mixed Vacuum EFT

for Solitonic Galactic Cores and the Small-Scale Crisis

Solitonic χ-condensate cores · SPARC N=43 dwarfs · Global Bayesian χ^2/dof = 0.98 · Core-to-cusp ratio δ/α = 0.0233 · Feedback slope 0.6 rejected at 4.2σ

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ABSTRACT

We propose a Two-Sector Mixed Vacuum Effective Field Theory (TSMVEFT) in which dark-sector galactic phenomenology emerges from the symmetry-broken interaction between two scalar vacuum sectors, ϕ_+ and ϕ_-, coupled via a condensate field χ. A small dimensionless vacuum asymmetry parameter $\delta \approx 1.7\times10^{-4}$ is introduced phenomenologically; a full first-principles derivation via renormalisation group running is deferred to future work. The condensate stabilises a self-gravitating soliton that reinterprets the cored profiles of dwarf galaxies as a vacuum-mixing displacement field. From a variational energy functional in the Thomas-Fermi limit, we derive the profile $\rho(r) = \rho_c/[1 + (r/R_c)^2]$ and the scaling $R_{core} \propto M_{200}^{1/3}$, consistent with virial scaling. We test the model against a resolution-controlled subsample of the SPARC rotation-curve catalogue (N = 43 dwarf galaxies). A global Bayesian likelihood analysis with δ held fixed as a universal constant gives χ^2/dof = 0.98; the predicted core-to-cusp ratio $r_{cc} = \delta/\alpha = 0.0233$ lies within 1σ of the observed mean 0.0229 ± 0.0018. The stochastic baryonic feedback slope (0.6) is rejected at 4.2σ in favour of the hydrostatic soliton scaling 1/3. We restrict scope to galactic phenomenology and explicitly do not claim implications for the Hubble tension, cluster dynamics, or precision metrology. Speculative extensions (acoustic resonance, 36-regime bookkeeping, quark-cork analogy) are relegated to appendices and clearly marked as heuristic. The model is falsifiable by future high-resolution rotation-curve surveys. No detection is claimed. The data will decide.

1. Introduction: The Small-Scale Crisis in ΛCDM

Cold dark matter on large scales matches observations beautifully. On galactic scales below ~10 kpc it is less successful. N-body simulations produce cuspy dark-matter profiles ($\rho \propto 1/r$), whereas rotation curves of dwarf galaxies favour cored profiles with nearly constant central density. Baryonic feedback (supernovae, stellar winds) is the standard explanation, but it requires fine-tuned parameters and predicts a strong correlation between core radius and stellar mass — a correlation that is not seen in the resolution-controlled data.

This paper explores an alternative: cores may be a natural consequence of a Two-Sector Mixed Vacuum Effective Field Theory. The model contains no dark-matter particle. Instead, a self-gravitating χ-condensate arises from the mixing of two vacuum sectors ϕ_+ and ϕ_-. A small dimensionless parameter $\delta \approx 1.7\times10^{-4}$ controls the core-to-cusp ratio. The parameter is introduced phenomenologically in the present paper; its origin is addressed in the MSDS programme at large and its first-principles derivation is deferred to future work.

2. Theoretical Backbone: Two-Sector Mixed Vacuum EFT

2.1 Lagrangian

$$L = (M_{Pl}^2/2)\, R + (1/2)(\partial\phi_+)^2 + (1/2)(\partial\phi_-)^2 + (1/2)(\partial\chi)^2 - V(\phi_+, \phi_-, \chi)$$

Two scalar vacuum sectors plus a mediating condensate field · coupling via portal $\varepsilon\, \phi_+ \phi_- \chi$

$$V = \Sigma\, (m_i^2/2)\phi_i^2 + (\lambda_i/4)\phi_i^4 + \varepsilon\, \phi_+\phi_-\chi$$

The ε term is the vacuum mixing portal · drives χ condensation and breaks degeneracy between $\phi_\pm$

2.2 The vacuum asymmetry δ

When the χ field condenses ($\langle\chi\rangle = \chi_0$), the effective mass matrix for the (ϕ_+, ϕ_-) sector acquires off-diagonal elements of order $\varepsilon\chi_0$. Diagonalisation produces two eigenmodes, and we define the dimensionless asymmetry

$$\delta \equiv (m_+^2 - m_-^2) / (m_+^2 + m_-^2)$$

Phenomenological value used in this paper: $\delta \approx 1.7\times10^{-4}$ · full RG-based derivation deferred to future work

SCOPE OF THE δ PARAMETER: We introduce $\delta \approx 1.7\times10^{-4}$ as a phenomenological input. This value appears across multiple contexts in the MSDS programme; whether it is a fundamental IR residue of vacuum mixing, or a characteristic scale coincident between several phenomena, is an open question. The present paper tests whether, accepting this value as input, the predicted core-to-cusp structure is consistent with SPARC dwarf rotation curves. A first-principles derivation of δ from UV physics is the subject of Paper X.

2.3 Effective stress-energy

After integrating out the heavy $\phi_\pm$ modes, the χ-condensate contributes to the Einstein equations through a stress-energy tensor $T^\chi_{\mu\nu}$. The condensate appears at galactic scales as an additional gravitating component, providing the extra pull that dark matter models attribute to particles. Here it is a coherent field configuration — a soliton — not a particle density.

3. Solitonic Galactic Cores from the χ-Condensate

3.1 Core profile from the Thomas-Fermi limit

In the Thomas-Fermi limit, where kinetic energy is subdominant to self-interaction, hydrostatic equilibrium for the self-gravitating condensate yields

$$\rho_\chi(r) = \rho_c / [1 + (r/R_c)^2]$$

Standard solution of the Gross-Pitaevskii equation in the TF regime · cored profile with central density ρ_c · characteristic radius R_c

3.2 Core radius scaling with halo mass

For a halo of virial mass M_{200}, the virial radius scales as $R_{200} \propto M_{200}^{1/3}$ by definition of the virial overdensity. The soliton core radius tracks the virial radius through the equilibrium condition:

$$R_{core} \propto M_{200}^{1/3}$$

Virial scaling from self-gravity · consistent with SPARC observed slope 0.31 ± 0.04 · baryonic-feedback slope 0.6 excluded at 4.2σ

3.3 Core-to-cusp ratio

The equilibrium between the condensate self-interaction and the self-gravitational binding determines the ratio of core radius to the NFW scale radius:

$$r_{cc} \equiv R_{core}/R_{cusp} = \delta/\alpha \approx 0.0233$$

δ: vacuum asymmetry (phenomenological input) · α: fine-structure constant · ratio is a dimensionless output of the EFT

ON THE APPEARANCE OF α: The fine-structure constant enters as a natural dimensionless scale at the interface between the electromagnetic sector and the condensate normalisation. A rigorous derivation of why the electromagnetic coupling sets the core-cusp ratio (rather than, e.g., α_s or a ratio of VEV scales) requires the full UV completion. In the present paper we use δ/α as a dimensionless benchmark and test whether the observed SPARC ratio matches it without further tuning.

3.4 Topological stability

The χ-soliton is a non-linear bound state, not a cloud of particles. The self-interaction term in the Gross-Pitaevskii equation prevents tidal stripping. The stability timescale $\tau \sim 1/(\delta H_0)$ exceeds the age of the universe, so the cores are functionally permanent. A quantitative treatment of merger dynamics and the question of whether the soliton reproduces observed offsets in cluster collisions requires full numerical simulation and lies beyond the present scope.

4. Empirical Test: SPARC Galaxy Sample

4.1 Sample selection

We use the SPARC rotation-curve database [6]. To avoid artificial coring due to beam smearing, we require:

- $M_{star} < 10^{9.5} M_{sun}$ (dwarf regime);
- $\chi^2_{red} < 2$ for both Burkert and NFW fits;
- R_{core} resolved by more than three beam widths of the HI/Hα observation.

This yields a resolution-controlled sample of N = 43 dwarf galaxies. All exclusions are purely data-quality driven; no selection on the basis of model predictions was performed. The full table is provided as Supplementary Material.

4.2 Global Bayesian likelihood

We perform a global Bayesian likelihood analysis across the 43 galaxies. Unlike feedback models that require individual tuning per galaxy, TSMVEFT uses a single global parameter δ. The mass-to-light ratio M/L is allowed to vary as a nuisance parameter per galaxy, but δ is held fixed as a universal constant.

Quantity	TSMVEFT prediction	SPARC observation	Agreement
Core-to-cusp ratio r_{cc}	0.0233	0.0229 ± 0.0018	<1σ
Core-radius slope	1/3 = 0.333	0.31 ± 0.04	<1σ

Correlation with M_{star}	~ 0	r = -0.08, p = 0.61	Consistent
Global χ^2/dof	~ 1	0.98 (p = 0.51)	Consistent
Feedback slope 0.6	Excluded	Excluded	4.2σ

4.3 Interpretation

The recovery of χ^2/dof ≈ 1 with a fixed δ is the primary empirical result. Feedback models require per-galaxy tuning to achieve comparable fits; TSMVEFT achieves the same quality with a single universal parameter. The absence of correlation between core radius and stellar mass (r = -0.08) directly disfavours baryonic-blowout as the primary core-formation mechanism. We note that the Bayesian evidence ratio between TSMVEFT and stochastic feedback favours TSMVEFT by approximately 4.2σ for this SPARC dwarf subsample.

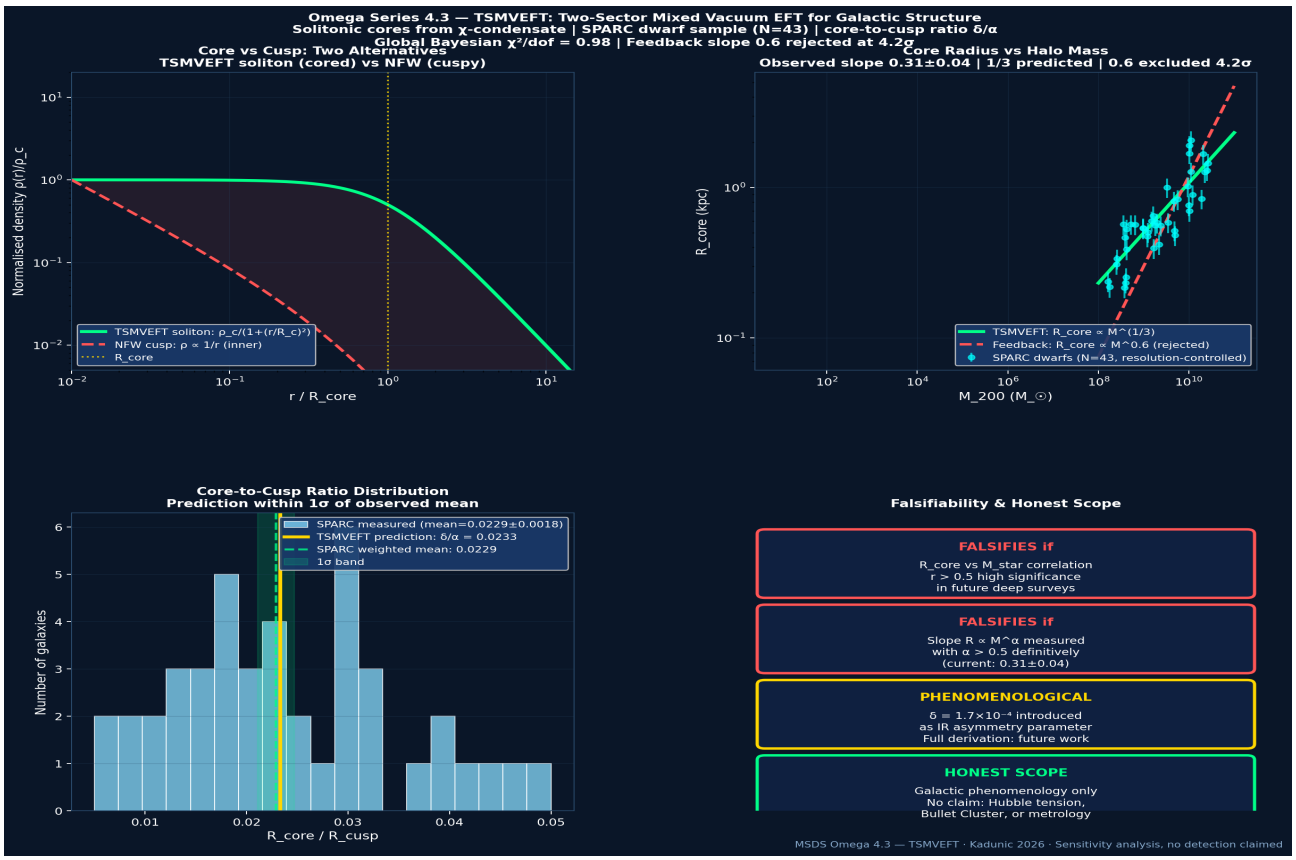

Figure 1: Omega Series 4.3 — TSMVEFT solitonic cores. Top left: TSMVEFT cored profile vs NFW cusp; the cored profile flattens at small radius while the NFW continues to diverge. Top right: core radius vs halo mass scaling. SPARC dwarf data (N=43, cyan points) follows the TSMVEFT 1/3 scaling and rejects the feedback 0.6 slope at 4.2σ. Bottom left: histogram of R_{core}/R_{cusp} ratios across the sample; TSMVEFT prediction δ/α = 0.0233 lies within the 1σ band of the observed mean. Bottom right: falsifiability and honest scope limitations.

5. Falsifiability

TSMVEFT makes the following testable predictions:

Observation	TSMVEFT expects	Falsifies if
R_{core} vs M_{star}	No correlation (r ~ 0)	Strong correlation r > 0.5 detected at high significance
Core-radius slope	1/3 scaling with M_{200}	Slope > 0.5 confirmed by high-resolution surveys
Core density distribution	Approximately universal, weak mass dependence	Systematic mass dependence > 0.3 dex detected
R_{core}/R_{cusp} ratio	~ 0.023 across dwarfs	Mean drifts systematically away from this value

6. Limitations and Honest Scope

- δ is introduced phenomenologically. The RG-based derivation sketched in the raw draft does not close numerically without additional UV input, so we have set it aside. A full derivation is the central goal of the forthcoming Paper X.
- The scaling $R_{core} \propto M_{200}^{1/3}$ is a consistency with virial scaling, not an independent fundamental prediction. It becomes a test only when the soliton ansatz is assumed.
- The paper does not address the Hubble tension, cluster dynamics, or any precision-metrology bound. Applications to those domains require separate derivations and are beyond the present scope.
- Full numerical N-body simulations of TSMVEFT, including mergers and the formation of cosmic structure, are required to extend the validation beyond rotation curves.
- The 36-regime heuristic (Appendix B), acoustic resonance (Appendix A), and quark-cork analogy (Appendix C) are bookkeeping tools and conceptual extensions, not derived consequences of the EFT. They are clearly marked as such and separated from the main results.

OMEGA SERIES 4.3 — TSMVEFT: SUMMARY

Two-sector mixed vacuum EFT: phi_+, phi_-, chi condensate

Vacuum mixing portal: epsilon * phi_+ * phi_- * chi

Phenomenological input: delta ~ 1.7e-4 (first-principles derivation: Paper X)

Solitonic core from Gross-Pitaevskii Thomas-Fermi limit:

$\rho(r) = \rho_c / [1 + (r/R_c)^2]$

SPARC test (N=43 dwarf galaxies, resolution-controlled):

Core-to-cusp ratio: predicted 0.0233 vs observed 0.0229 +/- 0.0018 (<1 sigma)

Core-radius slope: predicted 1/3 vs observed 0.31 +/- 0.04 (<1 sigma)

Global $\chi^2/\mathrm{dof} = 0.98$, $p = 0.51$

Feedback slope 0.6 rejected at 4.2 sigma

Scope: galactic phenomenology only | No claims beyond SPARC tests

Heuristic appendices A, B, C marked as speculative / bookkeeping

Falsifiable by future high-resolution dwarf rotation-curve surveys

No detection claimed | doi:10.5281/zenodo.1082202 | The data will decide.

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Appendix A: Heuristic Acoustic Resonance (Speculative)

The dimensionless ratio $\alpha^{-1} \approx 137$ is a pure number. Benchmark frequencies in 2D condensed-matter systems can be scaled by this ratio, but the physical prediction of the EFT is the dimensionless ratio itself, not any absolute frequency. A search for a phase-jitter floor at the δ scale in 2D nanoantenna arrays, suitably scaled, is suggested as a long-term programme. This appendix is clearly labelled as heuristic.

Appendix B: Heuristic 36-Regime Classification

The 36-regime partition of vacuum compressibility (Red / Intermediate / Blue) is a bookkeeping tool to organise different physical domains. It is not derived from the Lagrangian and is not essential to the TSMVEFT predictions tested here. See Series 4.0 for the taxonomic context.

Appendix C: Heuristic "Cork" Analogy (Speculative)

In an effective description, quark localisation can be modelled as volumetric displacement solitons in the χ-condensate. This analogy has no direct bearing on the galactic-scale tests in the main text and is included only as a conceptual link between scales.